HANIKA ANGLICAN INTEGRATED POLYTECHNIC (HAIP) · Data Mining and Data Warehousing · FA1 Group Work

# Team Contribution Report

Individual contributions, challenges faced and lessons learned

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Project: University Library Analytics Data Warehouse

Group: **Group X** (replace with your group number)

Members: Member 1 (Data Architect) · Member 2 (ETL Specialist) · Member 3 (Analytics & Security Lead)

Database: SQLite 3.53.1 · ETL: Python 3 + pandas · Dashboards: Power BI Desktop

Submission: November 2, 2026 · Presentation: November 6, 2026

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**To complete by the team:** replace *Member 1/2/3* with your names, adjust the percentages and dates, and rewrite the "personal reflection" lines in your own words. The tasks and challenges below are the ones of this project.

# 1. Team and roles

| Member | Role | Main responsibility |
|---|---|---|
| Member 1 - *name* | Data Architect & Database Designer | architecture, dimensional model, SQL scripts, data dictionary |
| Member 2 - *name* | ETL Specialist & Data Quality Manager | source data, ETL pipeline, cleansing rules, quality report, error log |
| Member 3 - *name* | Analytics, Reporting & Security Lead | business queries, OLAP, Power BI dashboards, RBAC, backup and recovery |

# 2. Individual contributions

## Member 1 - Data Architect

| Deliverable | Files |
|---|---|
| Kimball approach, 4-tier architecture, staging design | `01_Design/architecture.png`, Design Document ch. 2-3 |
| Galaxy schema (3 facts, 9 dimensions), ERD | `galaxy_schema.png`, `erd_full.png` |
| DDL scripts with keys, constraints, indexes, date dimension | `03_Database_Files/01-05_*.sql`, `build_database.py` |
| Data dictionary, schema justification, scalability, stakeholder answers | `01_Design_Document.pdf` |

Contribution: __ % · Hours: __

## Member 2 - ETL Specialist

| Deliverable | Files |
|---|---|
| Sample source data with all quality issues | `09_Source_Data/*` |
| ETL pipeline (extract, 30 cleansing rules, validation, transactional load) | `04_ETL_Files/etl_script.py` |
| Transformation rules, data quality report, error log, flowchart | `transformation_rules.xlsx`, `data_quality_report.pdf`, `error_log_sample.txt`, `etl_flowchart.png` |
| Nightly refresh job | `run_nightly_etl.bat` |

Contribution: __ % · Hours: __

## Member 3 - Analytics & Security Lead

| Deliverable | Files |
|---|---|
| 31 business queries (all 10 required + 3 bonus + extras) and results | `business_queries.sql`, `query_results.xlsx` |

| Deliverable | Files |
|---|---|
| OLAP demo, advanced analytics (correlation, forecast, anomaly) | `olap_operations_demo.pdf`, `advanced_analytics.py` |
| 3 Power BI dashboards, DAX measures, guide | `06_Dashboards/*` |
| RBAC, masking, row-level security, audit, backup and recovery | `06/07_*.sql`, `secure_access.py`, `backup_restore.py`, `07_Security_Documentation/*.pdf` |

Contribution: __ % · Hours: __

### Shared work

User Guide, presentation slides, testing (`run_all.py`), GitHub repository and final package.

## 3. Timeline

| Dates | Phase | Result |
|---|---|---|
| Oct 23-25 | Design | architecture, schema, DDL |
| Oct 26-27 | Database + sample data | working SQLite database, source files |
| Oct 27-29 | ETL | clean, loaded warehouse, quality report |
| Oct 29-31 | Analytics + security | queries, OLAP, dashboards, RBAC, backups |
| Nov 1-2 | Documentation + packaging | PDFs, slides, zip, GitHub |
| Nov 6 | Presentation | live demo |

## 4. Challenges faced and how we solved them

| Challenge | Solution |
|---|---|
| No real data was provided | wrote a generator that simulates the three systems with every issue from the brief (fixed seed, so all members get the same data) |
| "2024-05-03" means 5 March in one Excel sheet and 3 May elsewhere | date format chosen **per sheet** and the year checked against the sheet name |
| Near-duplicate bookings ("Morning"/"8AM-10AM", "R-101"/"Room101") | standardise first, then apply the business rule "one booking per room, day and slot" |
| Quarter-total rows with "1,203" hidden inside the Excel data | detected and rejected; otherwise downloads would be double-counted |
| SQLite has no users or GRANT | built `secure_access.py` with the SQLite authorizer: roles, column masking, own-department data, audit |
| Our first restore test failed (the restore skipped the differential backup; triggers flooded the audit log) | the automatic `verify` test found it; we fixed the restore chain and silenced triggers during restores - now every table passes |
| Too many loans counted as overdue with a 14-day rule | checked the distribution and adopted a 21-day loan period |

## 5. Lessons learned

- **Profile the data before cleaning it** - counting the different spellings and date shapes showed exactly which rules were needed.

- **Keep raw data in staging** - we could re-run the cleaning many times without going back to the sources.
- **Store rules as data** (mapping tables) - a new spelling needs one new row, not new code.
- **Test backups by restoring them** - a backup that has never been restored is only a hope.
- **One script to rebuild everything** (`run_all.py`) made team work and the demo reliable.
- *Personal reflection - Member 1:* ...
- *Personal reflection - Member 2:* ...
- *Personal reflection - Member 3:* ...